Group Project Report

What should the "you may also like" shelf show, and to whom?

Association rules · Lumen Beauty (Sep–Nov 2026)

Team number	Team 5
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1. Problem & what it is worth

After every purchase, Lumen Beauty's Douyin mini-program shows a *you may also like* cross-sell shelf. The brand founder and the e-commerce merchandising team must decide which association rules earn a shelf slot, and whether the shelf should be the same for first-time buyers as for frequent ones. Q15's business risk is not only a missed add-on; it is a shelf that quietly mirrors the heaviest buyers.

Across the cleaned quarter (1 Sep–30 Nov 2026) the store recorded **3,401 orders**, **647 buyers** and **¥1,171,735** GMV. **65%** of orders contain a single product (2,219 of 3,401), so most checkouts leave with nothing added. About **127 serum orders per month** leave without a cream (Sep 137 / Oct 124 / Nov 121). If the shelf persuades 5–15% of those orders to add a cream (average cream line ¥347), the scenario value is **¥2,206–6,619 per month** (0.6–1.7% of monthly GMV). That yuan figure is a scenario, not a measured lift. The sharper stake is who trains the shelf: heavy buyers (6+ orders) are 41% of buyers but place 80% of orders; light buyers (1–2 orders) are 46.5% of buyers but place only 11% of orders.

Getting product discovery wrong is not abstract. A shelf trained on histories would allocate scarce recommendation slots to pairs that heavy buyers assemble across many visits — pairs light buyers almost never create. That is a targeting error under STP: the store would position the shelf for its heaviest segment while nearly half of buyers are light. The opportunity cost of the empty second slot on 2,219 single-product orders is the scenario above; the fairness cost is who the algorithm listens to.

Unit of analysis (clarified). We mine rules on *two* basket definitions required by Q15: (1) single orders — the moment the shelf appears; (2) whole customer histories — every distinct product a buyer purchased in the quarter. The figure 1.45 is the average number of *distinct*

products per order (order lines collapsed to unique SKUs), not units shipped.

2. Data

We use two tables from the course project dataset (Lumen Beauty full Sep–Nov 2026 operations export; see *project_data_dictionary*): **lb_order_lines.csv** (4,920 rows before cleaning) and **lb_users.csv** (1,000 registered users). Auxiliary tables (streams, comments, follows) are excluded because basket co-purchase does not need them.

Cleaning choices

- **Duplicates:** 5 exact duplicate order lines removed (export errors). 4,915 lines remain. GMV after cleaning = ¥1,171,735 (the project plan's ¥1,173,482 still counted duplicates).
- **Missing values:** empty *stream_id* only on store orders (1,213 lines). Per the data dictionary this means "not applicable", not missing data.
- **Units:** qty ∈ {1,2}; unit_price ¥54–505. Store list price is unique per SKU; paid/list ratios sit in 0.55–1.15 (livestream promotions).
- **Outliers:** orders per buyer 1–20 (median 3). Boxplot fence = 21; nobody exceeds it. Heavy buyers are kept — they are the group Q15 asks about.
- **Join check:** all 647 buyers map to *lb_users*; *n_orders* matches the order table for every buyer.

Channels are **livestream** (2,579 orders) and **store** (822). Values such as search / short_video / ad belong to *users.first_channel* (arrival), not to where the order was placed — a correction from the project plan.

Buyer type	Buyers	Orders	% of buyers	% of orders
Light (1–2 orders)	301	366	46.5%	10.8%
Mid (3–5)	81	304	12.5%	8.9%
Heavy (6+)	265	2,731	41.0%	80.3%

Table 1. Buyer intensity segments used for Q15 (cut on n_orders, not spend).

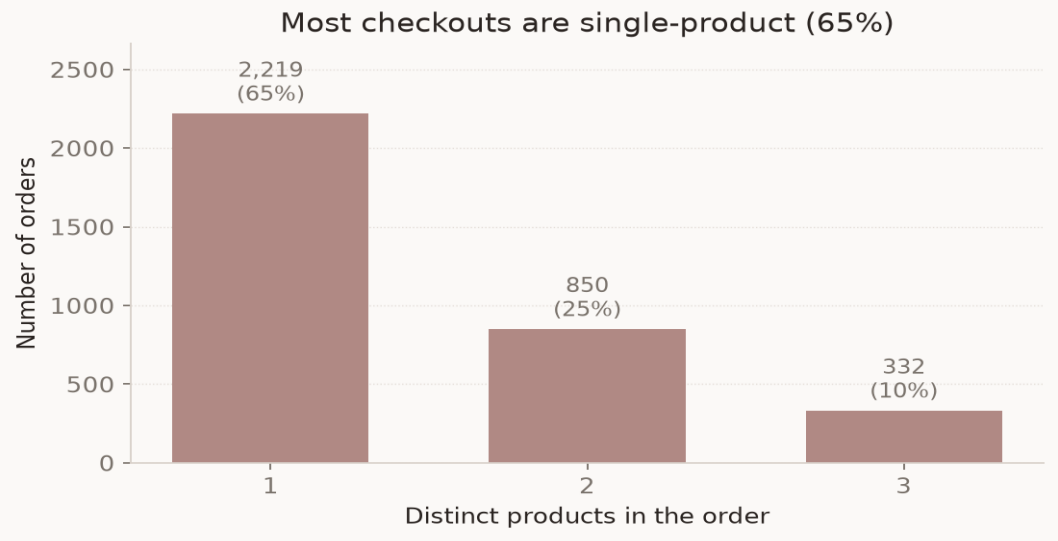


Figure A. Basket size: 65% of orders contain a single product — the shelf's opening.

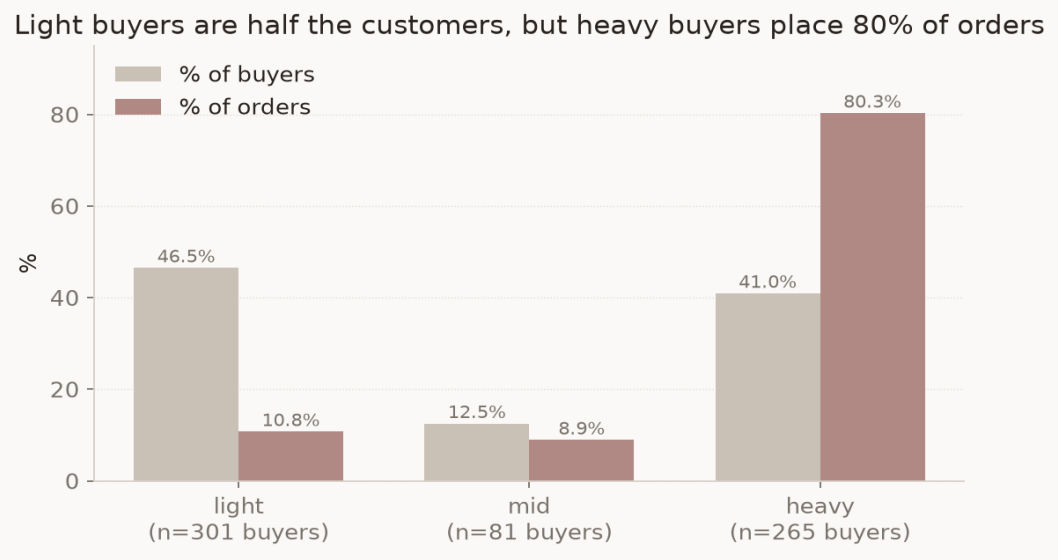


Figure B. Light buyers are half the customers; heavy buyers place 80% of orders.

Order baskets average **1.45 distinct products** (1 / 2 / 3 products: 2,219 / 850 / 332). History baskets average **6.27 products** — more than four times larger — which is why history mining floods the rule table.

Segmentation uses *n_orders* rather than *high_value* (spend-based), because Q15 asks whether frequent buyers dominate the rules — an intensity question, not a revenue-tier question. High-value flags still exist in *lb_users* (top 20% by spend, 130 users) but are not the cut that answers “to whom?” for this shelf. The chart that makes the business question visible is Figure 1 in §4: the gap between share of buyers and share of history-pair evidence.

3. Method & assumptions

Method family: **association rule mining** (Week 3), Apriori via *mlxtend*, pairs only (*max_len* = 2) because the shelf answers one product with one product. Metrics: support, confidence, lift. **Support and confidence filter observed co-purchase frequency; they are not tests of statistical significance.** Beside every rule we report the co-purchase count and its denominator (orders or customers).

Thresholds (decision-driven, not data-mined). The project plan's 2% product support returns zero product rules on orders (largest pair support = 0.79%). We set order-level *min_support* = **0.4%** (≥14 of 3,401 orders ≈ one qualifying order per week over 13 weeks). Category rules use 1% (≥35 orders). History rules use the plan's 2% (≥13 of 647 customers) so the two units of analysis face a comparable bar. The plan's 20% confidence filter is dropped for product rules (best confidence = 0.14); the category rule Serum → Cream still clears it (confidence 0.40).

Theory lens. Recommender algorithmic bias + STP targeting. Two testable predictions: (T1) if histories train the shelf, heavy buyers will dominate the evidence behind history rules; (T2) if one undifferentiated shelf is justified, Serum → Cream should hold for light, mid and heavy buyers and on first orders. A third effect — the live shelf creating the pairs it later mines — cannot be tested on past logs.

Assumptions & limits. Co-purchase is association, not causation. Past logs cannot measure incremental sales from a shelf that did not yet exist. Light buyers supply only 366 orders, so product-level rules inside that slice alone are thin; we trust the category rule more than SKU pairs.

How we read a rule

For every candidate rule A → B we require three numbers in plain language: how many baskets contain both (the numerator), how many baskets form the base (the denominator for support), and how many baskets contain A (the denominator for confidence). Lift then asks whether that confidence exceeds B's base rate. A rule can clear a support filter and still be useless for the shelf if lift ≤ 1, or if the pair is an artefact of large history baskets. That reading discipline is why Table 2 always shows both / denom beside support and confidence.

We deliberately avoid p-values on association rules. With thousands of possible pairs, significance testing without a pre-registered hypothesis would invite false discoveries. The shelf decision needs stable, interpretable co-purchase — which is why we demand a minimum weekly count and a lift above chance, then translate survivors into operational cards.

4. Findings

F1 · Order baskets vs history baskets

At a comparable bar (≥14 orders vs ≥13 customers), single orders yield **30 product rules**; whole histories yield **870** — every possible directed pair among 30 SKUs, all with lift > 1. Large history baskets (avg 6.27 products) manufacture pairs whether or not customers buy them together. A rule set that every pair passes cannot tell the shelf what to show.

This is the analytical heart of Q15. The same Apriori procedure, at almost the same minimum count, produces two incompatible shelves. The difference is not the algorithm; it is the basket definition. Histories answer “what has this customer ever tried?”; orders answer “what is bought together when the shelf appears?” Only the second matches the mini-program moment.

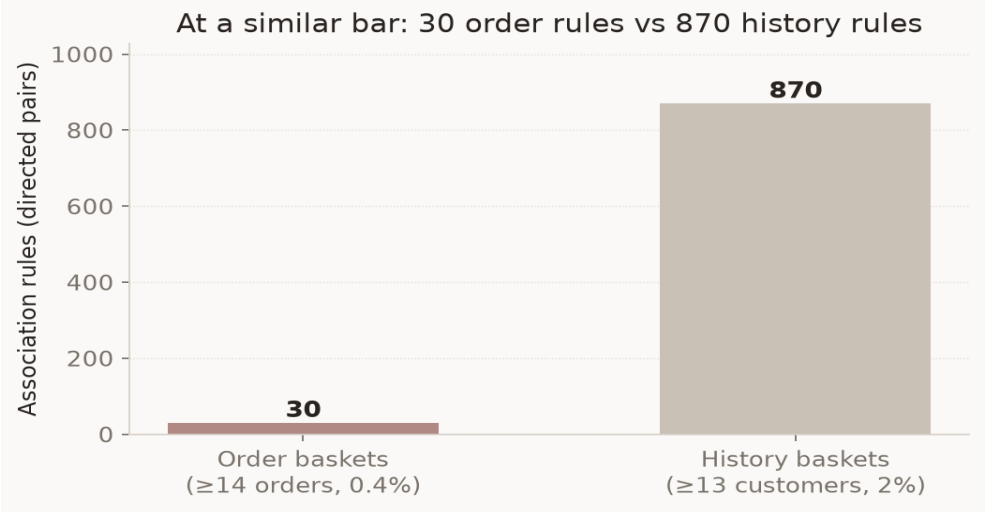


Figure E. At a comparable bar: 30 order rules vs 870 history rules.

F2 · The only strong order rule: Serum → Cream

At category level, **256 of 3,401 orders** contain both Serum and Cream against ~147 expected by chance (support 0.075, confidence 0.40, **lift 1.75**). Direction matters: Cream → Serum confidence is only 0.33, so the shelf should show a cream after a serum. Every other category pair at ≥1% support has lift < 1 (e.g. Lip–Eye Care 0.64). Product-level serum→cream pairs choose which cream to show first but rest on thin counts (15–27 orders).

Rule	Both / denom	Support	Conf.	Lift	Read as
Serum → Cream	256 / 3,401 orders (256 / 638 serum orders)	0.075	0.40	1.75	Show cream after serum
Vit. C Serum → Hydra Gel	26 / 3,401 (26 / 188 Vit. C)	0.0076	0.14	2.29	Cream #1 after Vit. C
Vit. C Serum → Brightening	27 / 3,401 (27 / 188 Vit. C)	0.0079	0.14	2.27	Cream #2 after Vit. C

Table 2. Selected order rules with co-purchase counts and denominators (filters = observed frequency, not significance tests).

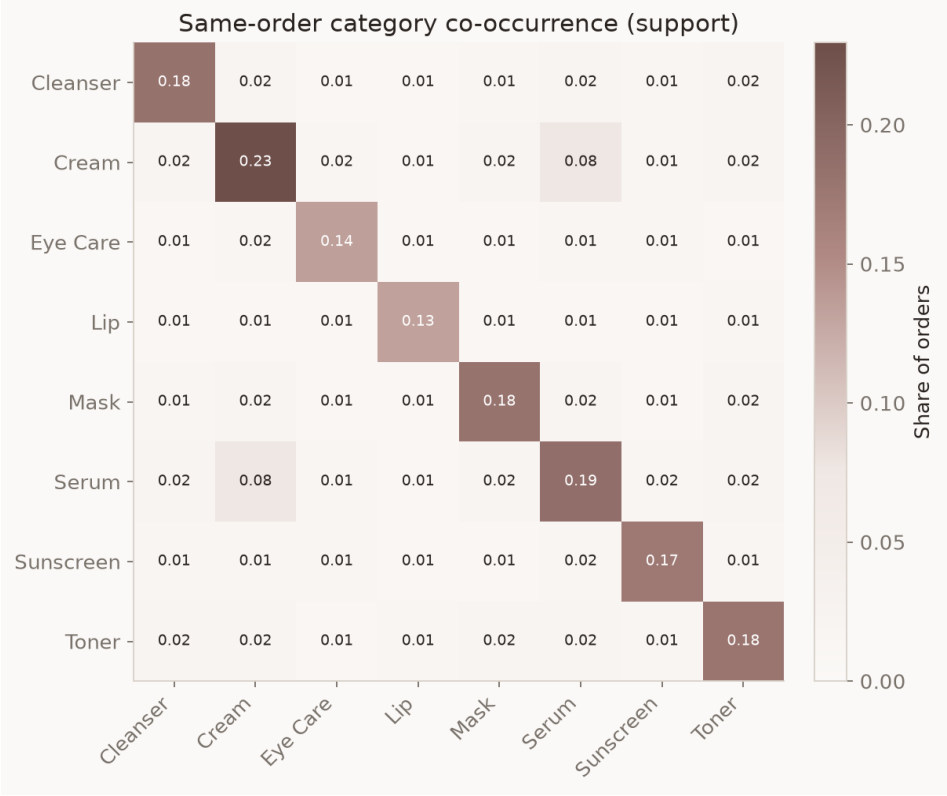


Figure C. Same-order category co-occurrence (support). Serum ∩ Cream = 0.075 stands out.

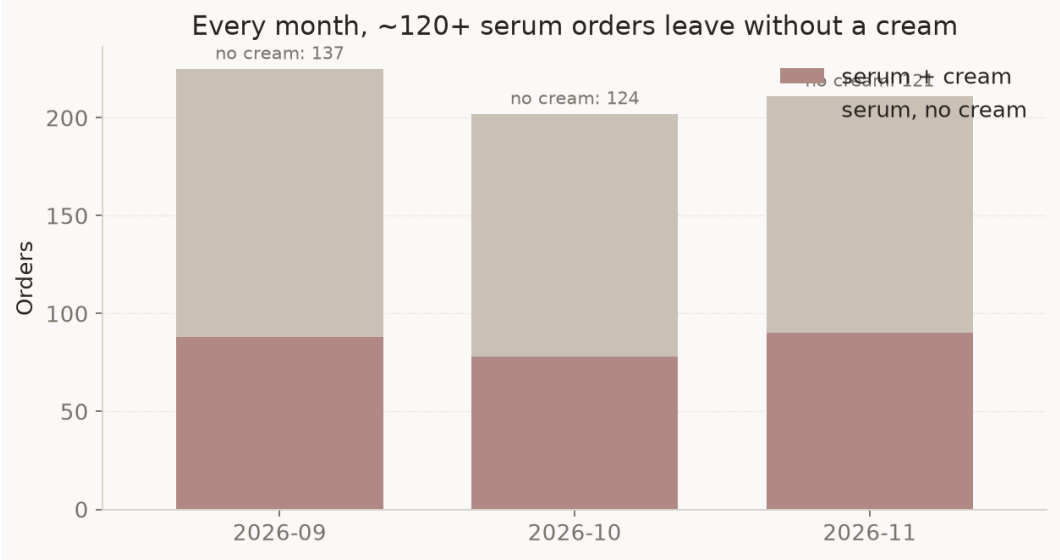


Figure D. Monthly serum orders with vs without a cream — the shelf's addressable set.

Hand check (Vitamin C Serum → Hydra Gel Cream): 26 of 3,401 orders contain both (support 0.0076); 26 of 188 Vitamin C orders also contain the cream (confidence 0.14); chance predicts ~11 joint orders, we observe 26 (lift 2.29). The numbers from *mine()* match the hand count, so the pipeline is faithful to Week 3. Because product confidence tops out at 0.14, we do *not* treat SKU pairs as standalone shelf engines — only as a ranking layer under

the category rule.

Going beyond the menu question: we also re-read Serum → Cream inside livestream vs store orders. Lift remains above 1.7 in both. Unlike host-manufactured bundles that appear only on stream, this habit survives on the store shelf, so it is a customer pattern rather than a presenter artefact.

F3 · History rules are written by heavy buyers

A history with *n* products creates *n(n–1)/2* pairs. Heavy buyers average 11.7 products and supply **92.9%** of all pair evidence; light buyers are 46.5% of buyers but only **1.7%** of the evidence. Behind each of the top five history rules, 90–95% of backing customers are heavy. Robustness: at every heavy cut-off (4+, 6+, 8+, 10+ orders), heavy buyers' share of evidence is two to three times their share of buyers (even at 10+: 21% of buyers, 63% of evidence).

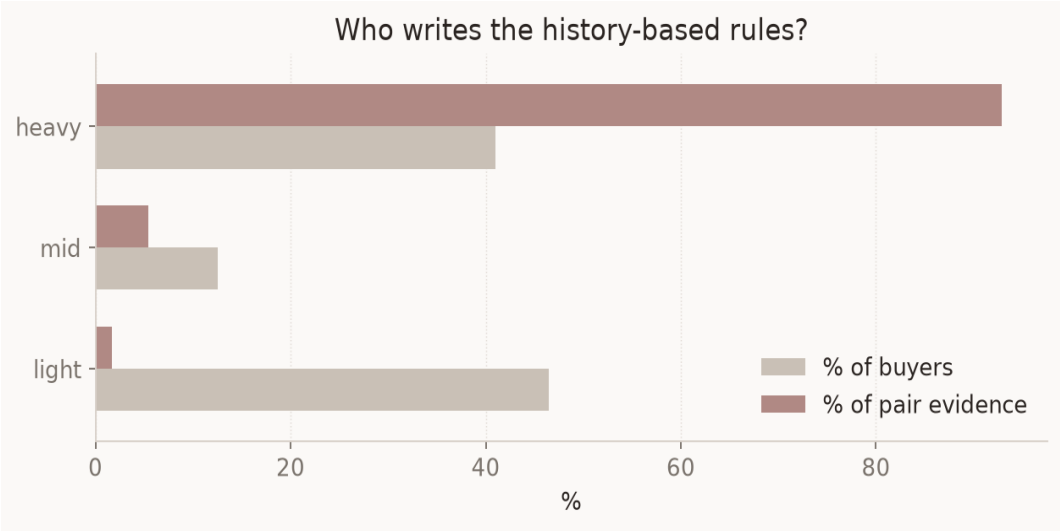


Figure 1. Who writes the history-based rules? Share of buyers vs share of pair evidence, by buyer type.

Heavy cut-off	% of buyers	% of history-rule evidence	Serum→Cream lift (heavy)
4+ orders	48.2%	97.1%	1.73
6+ (main)	41.0%	92.9%	1.72
8+ orders	32.8%	83.3%	1.71
10+ orders	21.2%	62.8%	1.61

Table 3. The heavy-buyer dominance finding does not depend on the cut-off.

F4 · Top history pairs are not bought in the same order

Inside single orders, the top history pairs have **lift < 1**. Calming Toner Mist → Pore Refining Toner has history lift 2.15 but same-order lift 0.37 (3 orders with both vs ~8 expected). Heavy buyers try substitutes across visits; labelling those pairs "frequently bought together" would misdescribe the data.

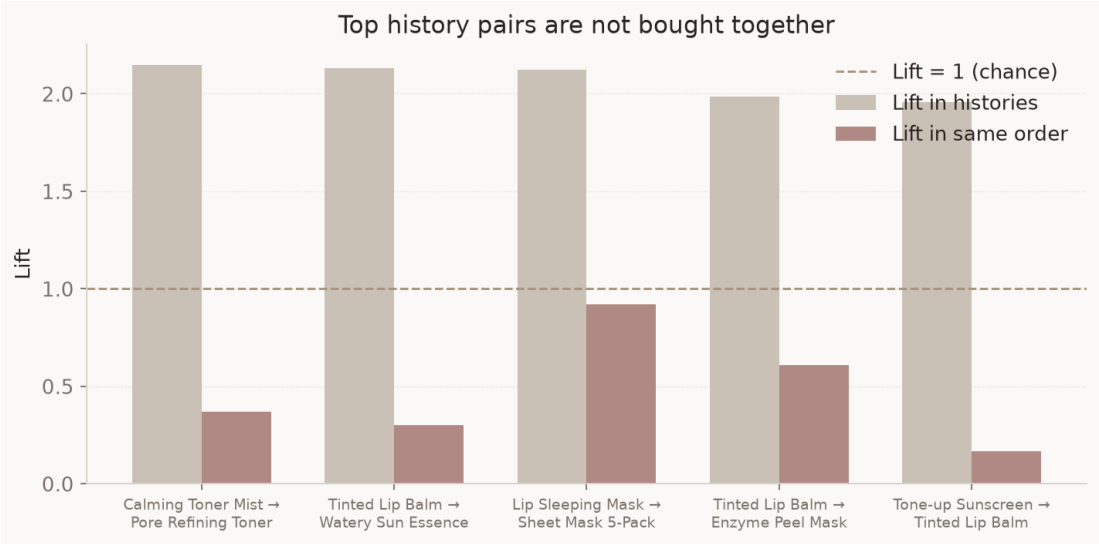


Figure F. Top history pairs: high history lift, same-order lift below 1.

F5 · One shelf serves new and occasional buyers — if order-based

Serum → Cream holds for light / mid / heavy (lift 1.79 / 1.93 / 1.72), livestream and store (1.73 / 1.80), and on each customer's **first order** (confidence 0.43, lift 1.80). A separate first-time shelf is unnecessary *provided* the shelf is mined from single orders.

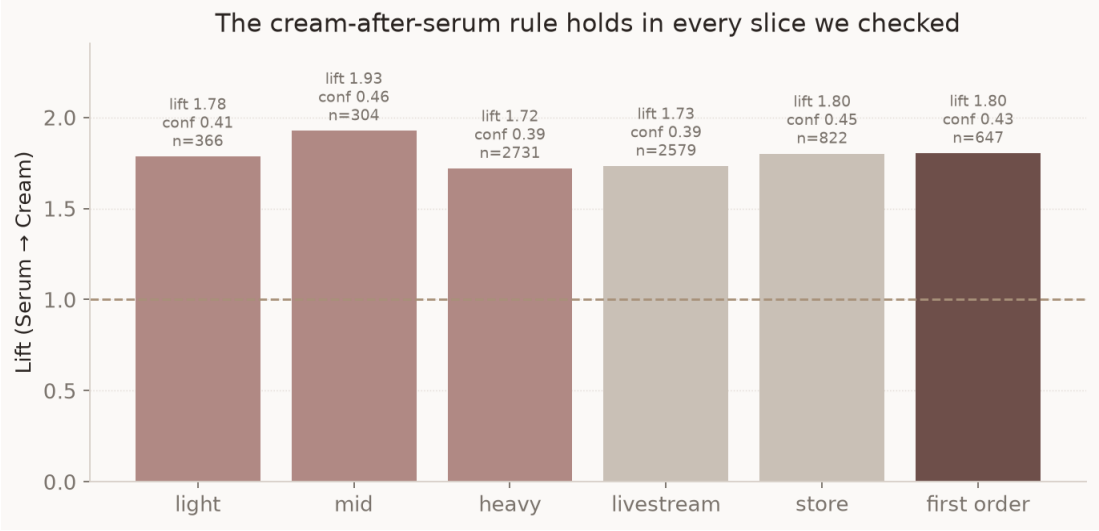


Figure G. Serum → Cream lift across buyer types, channels, and first orders.

Slice	Orders	Support	Confidence	Lift
buyer_type = light	366	0.093	0.410	1.79
buyer_type = mid	304	0.086	0.456	1.93
buyer_type = heavy	2,731	0.072	0.394	1.72
channel = livestream	2,579	0.074	0.387	1.73
channel = store	822	0.080	0.449	1.80
first order only	647	0.085	0.426	1.80

Table 4. Serum → Cream across slices.

Stretch (Q15). Mining category rules separately on light vs heavy *order* baskets still returns Serum → Cream as the primary rule in both poles, with lift well above 1. The stretch therefore rejects two product shelves; it reinforces one order-based cream-after-serum shelf.

What would change our mind before the pilot. If a later month showed Serum → Cream same-order lift ≤ 1, or if light-buyer confidence collapsed while heavy-buyer confidence held, we would stop claiming one shelf for everyone. Neither appears in this quarter.

F6 · What this means for the business argument

Three questions decide whether an NMTA project succeeds: what the question is worth, what the data say, and what someone should do differently on Monday morning. The data say that the history-trained shelf fails the "to whom?" part of Q15, while the order-trained Serum → Cream rule passes every slice we could run — including buyers with no history. The yuan scenario is modest; the decision that matters is which evidence base trains the shelf.

Relative to the project plan (20 Oct 2026), three corrections matter. First, baskets are built two ways, not only by order_id. Second, segmentation uses order intensity, not high_value spend. Third, thresholds are justified by the shelf decision after the plan’s 2% / 20% filters returned no usable product rules. Support is never described as "statistically meaningful": it is a frequency filter, and every shipped rule carries its count and denominator.

Relative to the instructor email: expected results and the drop condition are stated in §5; shelf decision cards name trigger, customers, position, the no-history view, and reject evidence; AI tools are named in the appendix. We go beyond the menu by comparing livestream vs store, first orders, and the light-vs-heavy stretch shelf.

5. Recommendation

Monday-morning decision. Mine the post-purchase shelf from **single orders**, not from whole histories. After any serum purchase, show a cream. Use thin product pairs only to rank which cream appears first. One shelf for everyone — including buyers with no prior history. Fill remaining slots with products light buyers buy most, not overall bestsellers (bestsellers are 80% written by heavy buyers’ order volume).

Operationally: (1) switch mining from user-history baskets to order baskets; (2) hard-code Serum → Cream as slot 1 whenever the basket contains a serum and no cream; (3) rank cream SKUs by order-level lift among serum→cream pairs with ≥14 supporting orders; (4) fill leftover slots from the light-buyer frequency list refreshed weekly.

Shelf decision card (selected rules)

Element	Serum → Cream (primary)	Vit. C → Hydra Gel (SKU rank)
Trigger product	Any serum in the current basket	Vitamin C Serum in basket
Intended customers	All buyers, including first-time / light	Serum buyers choosing among creams
Shelf position	Slot 1 on post-purchase shelf	Slot 1 among cream candidates after Vit. C
Buyer with no history sees	Same cream recommendation (first-order lift 1.80)	Same ranked cream if Vit. C was just bought
Evidence that rejects the rule	Same-order lift ≤1 in a holdout month, or A/B add-on < 2% for six weeks	Co-purchase count < 10 / quarter or lift ≤1 inside orders

Table 5. Translating rules into shelf decisions.

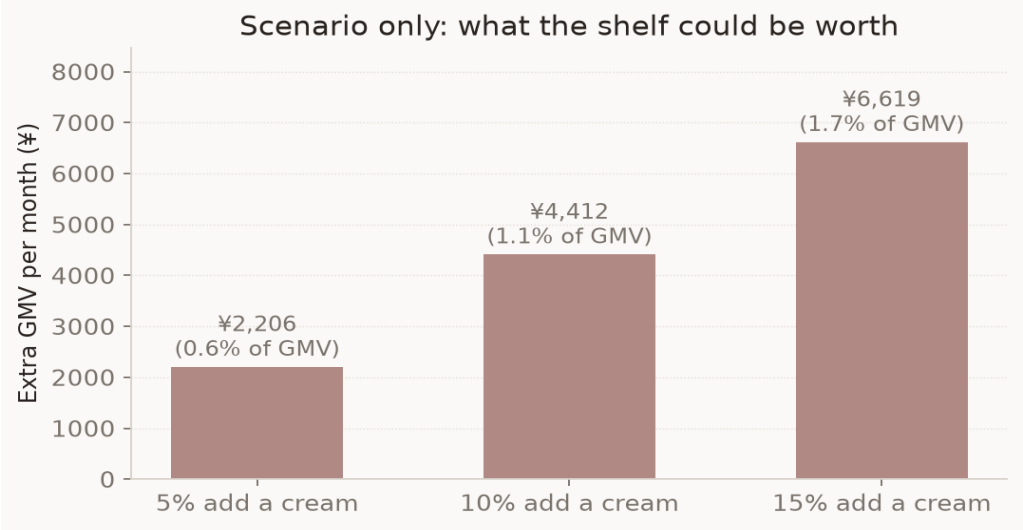


Figure H. Scenario value if 5–15% of serum-only orders add a cream (not a measured lift).

Expected results & condition that drops the action

- **Pilot:** from 1 Dec 2026, show the new shelf to a random half of buyers for six weeks; keep the current shelf for the other half.
- **Expected result:** ≥5% of serum-only orders in the treatment half add a cream (~¥2,200 / month at the low end), reported separately for first-time and repeat buyers.
- **Drop condition:** stop if (a) six-week cream add-on rate in treatment is < 2% and not higher than control, or (b) light / first-time add-on rate is materially below heavy buyers', or (c) a re-mine on the control half shows Serum → Cream same-order lift ≤ 1.

Where the recommendation stops

There is no second strong category pair. Do not invent a multi-rule carousel from history lifts. Do not personalise by full purchase history for new buyers. Do not claim the ¥2.2–6.6k figure as a measured effect. The pilot is the measurement. If the kill conditions fire, revert to the prior shelf and re-open the analysis with the next quarter's orders — still on order baskets, not histories.

Ethics (§6) and AI-use disclosure (§7) are in the separate appendix, per the instructor's 12 November formatting note.